Arista Access Node (MAR-ACC-04) Optical Transport Failure SOP (ALARM_7586)

Node: MAR-ACC-04 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_7586

Description

This Standard Operating Procedure (SOP) addresses the critical failure scenario of Optical Transport Failure on Arista Access Node (MAR-ACC-04) located at the Marathahalli Core site, characterized by an Alarm Code of ALARM_7586. The failure domain encompasses SFP/XFP degradation, fiber cuts, and Rx/Tx power loss, compromising the integrity of the access network. The underlying protocol or hardware mechanics responsible for this failure domain are rooted in the Arista EOS (Extensible Operating System) software's reliance on Optical Transport Network (OTN) protocols, specifically the G.709 framing and Forward Error Correction (FEC) mechanisms. In this scenario, the degradation of SFP/XFP modules or the occurrence of fiber cuts results in a failure to maintain the requisite signal quality and integrity, leading to the generation of the ALARM_7586 alarm. The severity of this failure is classified as Critical, necessitating swift and accurate corrective action.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Marathahalli Core site to provide physical access to the MAR-ACC-04 node.

- Verify the status of redundant links to determine if the failure has impacted network connectivity.

- Check the environment for any signs of physical damage or tampering.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface | include loss, Expected Output: Rx power loss detected, Description: This command isolates the Rx power loss issue by filtering the output to display only the relevant information, allowing for the identification of the affected interface.

Command: show logging | include ALARM_7586, Expected Output: ALARM_7586 generated due to SFP/XFP degradation, Description: This command analyzes the system logs to determine the root cause of the ALARM_7586 alarm, providing insight into the nature of the failure.
```

Resolution Steps

Step 1: Step: Replace SFP/XFP module, Command: replace module with , Description: This step involves replacing the degraded SFP/XFP module with a new one, ensuring the integrity of the optical signal and restoring network connectivity.

Step 2: Step: Verify fiber integrity, Command: show interface | include fiber, Description: This step entails verifying the physical integrity of the fiber by checking for any signs of damage or tampering, ensuring a secure and reliable optical connection.

Post-Check Verification

```
Command: show interface | include status, Expected Output: Interface up and running, Description: This command verifies the status of the interface, confirming that it is up and running, and the issue has been resolved.

Command: show logging | include ALARM_7586, Expected Output: ALARM_7586 cleared, Description: This command ensures that the ALARM_7586 alarm has been cleared, indicating that the failure has been successfully resolved.
```

Roll-back Procedures

- Step: Restore original SFP/XFP module, Command: replace module with , Description: In the event of a failed fix, this step involves restoring the original SFP/XFP module, ensuring the integrity of the optical signal and network connectivity.

- Step: Verify original fiber integrity, Command: show interface | include fiber, Description: This step entails verifying the physical integrity of the original fiber, ensuring a secure and reliable optical connection.